■ 1.8.9 Advanced Topic: Shading Surfaces with Functions

The last section discussed how you can shade surfaces either according to their height, or according to the effects of simulated illumination.

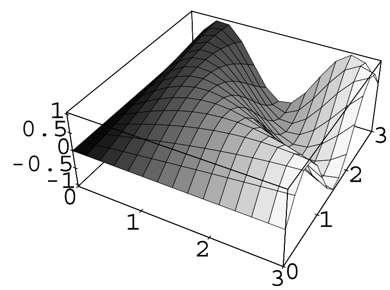
With *Mathematica*, you can also specify explicitly how each part of a surface is to be shaded.

`Plot3D[{`*f*`,` *s*`}, {`*x*`,` *xmin*`,` *xmax*`}, {`*y*`,` *ymin*`,` *ymax*`}]`	plot a surface corresponding to *f*, shaded according to the function *s*

Specifying shading functions for surfaces.

This shows a surface whose height is determined by the function `Sin[x y]`, but whose shading is determined by `GrayLevel[x/3]`.

```
In[1]:= Plot3D[{Sin[x y], GrayLevel[x/3]},
                          {x, 0, 3}, {y, 0, 3}]
```



You can use either `GrayLevel` or `RGBColor` to specify a shading function for a surface. You can get many different effects by choosing different shading functions. If you effectively want to display an extra coordinate, you can often do this using a color shading function.